

Is there any difference between hd3ss322irnhr and hd3ss3212 whats these

Both HD3SS3220 (full orderable part number HD3SS3220IRNHR) and HD3SS3212 are Texas Instruments integrated circuits used for routing high-speed differential signals up to 10 Gbps (USB 3.2 Gen 2, PCIe Gen 3, DisplayPort). The main difference is that the HD3SS3220 is an all-in-one USB Type-C port controller with an integrated switch, while the HD3SS3212 is purely a standalone passive switch IC.

Feature	HD3SS3220 (HD3SS3220IRNHR)	HD3SS3212
Primary Function	USB Type-C Port Controller + SuperSpeed MUX	Standalone 2-Channel 2:1 / 1:2 Differential MUX
Type-C CC Logic	Built-in (Detects plug orientation, DFP/UFP/DRP roles)	None (Requires an external CC controller or MCU)
Switch Control	Automatic (internal CC logic flips lines based on plug direction)	Manual (controlled externally via a SEL pin)
Supported Protocols	USB 3.1 / 3.2 SuperSpeed (Type-C applications)	USB 3.1, PCIe Gen 3, DisplayPort, SATA
Package / Order Code	30-pin WQFN (RNH package, Industrial temp I)	20-pin WQFN / 18-pin X2QFN
Design Complexity	Low (Single-chip solution for Type-C ports)	Flexible (Requires external control logic)

HD3SS3220IRNHR

Autonomous CC Logic: Automatically handles Type-C Configuration Channel (CC) functions including cable insertion detection, orientation detection, and role configuration (DFP/Host, UFP/Device, or DRP/Dual Role).

Automatic Routing: Dynamically routes the USB 3.x SuperSpeed TX/RX lanes to match whichever way the user plugs in the Type-C cable.

Best Used For: Adding a standard USB Type-C port to a hardware design with minimal component count and no extra firmware requirements.

HD3SS3212

Passive MUX/DEMUX: Operates strictly as a high-bandwidth physical switch for differential pairs.

Protocol Agnostic: Routes any high-speed differential signals up to 10 Gbps regardless of protocol.

External Control: Requires an external pin signal (from an MCU, FPGA, or dedicated CC controller like the TUSB320) to toggle the signal routing path.

Best Used For: Custom high-speed signal switching, multi-lane PCB routing, or Type-C designs where a separate host controller manages the CC logic.

I am going to design audio over usb its like nowadays, new smartphones having single usb type c port for charging, audio and data transfer in orcad capture id this block diagram is well suited for this requiremrnt and how to do it in orcad capture for every ic pins

This block diagram is very well suited for a multi-function USB Type-C implementation supporting charging, USB 2.0/3.x data, and analog audio accessory mode.

Architecture & Block Diagram Evaluation

Strengths:

TS5USBA224: Excellent choice for the USB 2.0 / Audio switch. It is specifically designed with negative-rail capability (down to -2.8V) so audio AC signals pass through without distortion.

HD3SS3212: Industry-standard passive high-speed 2:1 switch for routing SuperSpeed USB 3.x (TX/RX) lanes according to cable insertion direction.

TUSB320: Reliable CC logic controller for plug orientation detection, role negotiations, and driving the MUX select pins.

MAX77751: Capable autonomous battery charger with integrated power-path management and 16V protection capability.

Key Additions / Considerations:

Microphone & Ground Switching (SBU Pins): If you intend to support standard 4-pole TRRS headsets (with a microphone), you must also route the SBU1 and SBU2 lines from the Type-C port through an aux audio switch (such as the TI TS3A226AE or TS3A227E) to automatically switch MIC and GND lines.

CC Line Loading: Both the MAX77751 and TUSB320 feature CC detection. Ensure TUSB320 handles primary CC state/orientation detection and mode switching, while MAX77751 is configured strictly for VBUS charger current detection to avoid conflicting pull-down/pull-up resistors on the CC lines.

Interconnection & Pin Mapping Guide

1. USB Type-C Receptacle Interconnects

VBUS (Pins A4, A9, B4, B9): Short together \rightarrow Route to OVP IC Input \rightarrow MAX77751 (CHGIN).

GND (Pins A1, A12, B1, B12): Connect to system Ground plane.

CC1 (A5) & CC2 (B5): Route directly to TUSB320 (CC1, CC2).

D+ (A6, B6) & D- (A7, B7): Short A6+B6 together as USB_C_D+ and A7+B7 as USB_C_D- \rightarrow Connect to TS5USBA224 (D+, D- Common pins).

TX1+/-, TX2+/-, RX1+/-, RX2+/-: Route as 90\ \Omega differential pairs to HD3SS3212 port B and port C pins.

SBU1 (A8) & SBU2 (B8): Route to Audio CODEC / MIC switch if headset microphone support is

required.

2. TUSB320 (CC Logic Controller)

CC1, CC2: Inputs from USB-C Port pins A5 and B5.

DIR Output: Connect to HD3SS3212 (SEL) to toggle SuperSpeed lanes based on cable orientation.

OUT1 / OUT2 or AUDIO_ACC_N: Connect to TS5USBA224 (ASEL / Control Pin) to switch D+/D- from USB 2.0 Data mode to Audio Mode when an audio dongle is detected.

I2C_SDA / I2C_SCL: Connect to Main Processor I2C bus (optional if operating in autonomous pin mode).

3. TS5USBA224 (USB 2.0 / Analog Audio Switch)

D+ / D- (Common): Connected to USB-C Receptacle D+ / D-.

USB_DP / USB_DM (Data Path): Connect to Main Processor USB 2.0 Data controller.

R_OUT / L_OUT (Audio Path): Connect to Audio CODEC / DAC Output (L and R analog lines).

ASEL (Select Pin): Driven by TUSB320 or Processor GPIO (HIGH = Audio, LOW = USB Data).

4. HD3SS3212 (SuperSpeed USB 3.x 2:1 MUX)

Port A (Common Differential Pairs): Connect to Main Processor USB 3.0 SuperSpeed SSTX and SSRX pins.

Port B & Port C: Connect to USB-C receptacle TX1/RX1 and TX2/RX2 pairs respectively.

SEL: Driven by TUSB320 (DIR).

OEN (Output Enable): Tie to Ground (LOW) for active operation.

5. MAX77751 & OVP (Power & Battery Charger)

CHGIN: Connect to Output of OVP IC (derived from USB-C VBUS).

SYS: Main System Power rail (powers PMIC / regulators).

BATT: Connect to Battery positive terminal (+).

STAT / INOKB: Connect to Processor GPIO interrupts or LED indicators.

Step-by-Step Implementation in OrCAD Capture

Symbol Creation (Schematic Library):

Create or download OrCAD Capture symbols (.OLB) for TUSB320, HD3SS3212, TS5USBA224, and MAX77751. Ensure pin names match the datasheet exact pin numbers and types (Input, Output, Power, Passive, Bidirectional).

Hierarchical or Multi-Page Schematic Setup:

Page 1: Connector & Power Protection: USB-C Receptacle, ESD Protection Diodes, OVP IC, and MAX77751 Charger.

Page 2: Type-C Control & Multiplexing: TUSB320 Controller, HD3SS3212 SuperSpeed MUX, and TS5USBA224 Audio Switch.

Page 3: Audio Codec & Main Processor: Audio DAC/Codec and Processor USB interfaces.

Wiring & Differential Pair Net Naming:

Use Off-Page Connectors (OFFPAGELEFT-L / OFFPAGERIGHT-R) to link signals between schematic pages cleanly.

Assign proper differential pair names to high-speed routes in OrCAD Capture (e.g., USB_SS_TX_P, USB_SS_TX_N) to prepare for PCB Layout rules.

Design Rules Check (DRC):

Go to Tools \rightarrow Design Rules Check in OrCAD Capture.

Run DRC to verify there are no duplicate pin numbers, floating input pins, unassigned net names, or shorted power rails before generating the netlist for PCB design.

Ovp and pmic is not using because max77751 has built in ovp and all

That is completely correct. The MAX77751 is an autonomous 3.15A USB Type-C buck charger with built-in power-path management and over-voltage protection, making both a discrete OVP IC and a separate charger PMIC unnecessary.

Built-In MAX77751 Capabilities

Integrated OVP Protection: Withstands up to 16V input transients and features integrated Over-Voltage Protection on its CHGIN pin (13.7V max input operating threshold).

Smart Power Selector™ (Power-Path Management): Automatically directs power from VBUS to the system rail (SYS) while simultaneously charging the battery (BATT), or routes power from the battery to SYS when VBUS is disconnected.

Integrated BC1.2 & CC Sockets: Detects standard legacy USB chargers (SDP, CDP, DCP) and Type-C current capability automatically.

Updated Schematic Pin Connections for OrCAD Capture

VBUS Power Input Line:

USB Type-C Port VBUS (Pins A4, A9, B4, B9): Short together \rightarrow Connect directly to MAX77751 (CHGIN).

Protection Requirement: Place a high-speed ESD/TVS diode array (e.g., 5V TVS) and decoupling capacitors (10 μ F + 0.1 μ F) near the CHGIN pin to clamp ESD strikes.

System Power & Battery Pins:

CHGIN: Direct input from USB VBUS.

SYS: System supply output rail. Connect this pin to downstream LDOs / Buck converters powering your audio CODEC, processor, and switch ICs.

BATT: Connect directly to the positive (+) terminal of the 1S Li-Ion / Li-Po battery.

INOKB: Connect to a status LED or Processor GPIO (pulls LOW when a valid VBUS source is present).

Shared CC Pin Wiring:

CC1 (A5) & CC2 (B5) Lines: Connect in parallel to TUSB320 (CC1, CC2) and MAX77751 (CC1,

CC2).

Control Hierarchy: TUSB320 manages cable orientation output signals (DIR to HD3SS3212 and OUT1/OUT2 to TS5USBA224), while MAX77751 uses CC lines purely for VBUS current limit detection.

Verification Step

In OrCAD Capture, run Design Rules Check (Tools \rightarrow Design Rules Check) to verify that VBUS connects directly to MAX77751.CHGIN as an POWER_INPUT without throwing floating pin errors, and confirm MAX77751.SYS is marked as POWER_OUTPUT feeding downstream power nets.

I need to know every ics pins where to connect and what to connect for good design

To build a reliable PCB design for USB charging, USB 3.x/2.0 data, and analog audio, follow this precise pin-by-pin connection specification and component setup for each IC in OrCAD Capture.

1. USB Type-C Receptacle Pins

VBUS (Pins A4, A9, B4, B9): Tie all 4 pins together to form the VBUS net \rightarrow Connect directly to MAX77751.CHGIN. Place a 10 μ F / 25V X7R ceramic capacitor, a 0.1 μ F cap, and a 5V TVS diode (e.g., ESD5B5.0ST5G) right at the connector pins.

GND (Pins A1, A12, B1, B12): Tie together to the main PCB Ground plane.

CC1 (A5) & CC2 (B5): Route CC1 to TUSB320.CC1 + MAX77751.CC1. Route CC2 to TUSB320.CC2 + MAX77751.CC2.

D+ (A6, B6) & D- (A7, B7): Short A6+B6 to form USB_C_DP \rightarrow Connect to TS5USBA224.COM1. Short A7+B7 to form USB_C_DM \rightarrow Connect to TS5USBA224.COM2.

TX1+/TX1- (A2/A3) & RX1+/RX1- (B11/B10): Route as 90 Ω differential pairs \rightarrow Connect to HD3SS3212.Port_B.

TX2+/TX2- (B2/B3) & RX2+/RX2- (A11/A10): Route as 90 Ω differential pairs \rightarrow Connect to HD3SS3212.Port_C.

Verification: Measure impedance from D+/D- and SS lines to GND to verify ESD diodes clamp properly without shorting lines.

2. TUSB320 (Type-C Controller & Cable Orientation Logic)

CC1 / CC2 (Pins 1 & 12): Connect to USB-C Receptacle CC1 and CC2.

VDD (Pin 4): Connect to MAX77751.SYS (3.3V rail). Place a 0.1 μ F bypass capacitor close to the pin.

GND (Pin 7 / EP): Connect to Ground plane.

DIR (Pin 6): Connect directly to HD3SS3212.SEL. Signals cable orientation (0 = \text{Normal}, 1 =

\text{Flipped}).

OUT1 / OUT2 (Pins 8 & 9): Connect to Processor GPIO or TS5USBA224.SEL to indicate when an audio accessory adapter or standard charger is attached.

PORT (Pin 2): Connect to VDD through a 10 k Ω resistor to set Dual Role Port (DRP), or to GND for Unattached/UFP.

I2C_DISABLE / ADDR (Pin 3): Tie to GND to enable I2C mode.

SDA / SCL (Pins 10 & 11): Connect to Processor I2C bus with 4.7 k Ω pull-up resistors to 1.8V/3.3V V_{DDIO} .

Verification: Toggle cable insertion direction; verify DIR pin state flips between Logic 0 and Logic 1.

3. HD3SS3212 (SuperSpeed 10 Gbps Differential MUX)

VDD (Pins 4, 9, 15): Connect to 3.3V supply. Place a 0.1 μ F ceramic capacitor at every VDD pin.

GND (Pins 5, 10, 16 / EP): Connect directly to Ground.

SEL (Pin 2): Connect to TUSB320.DIR.

OEN (Pin 1 - Output Enable): Tie directly to GND to keep the multiplexer active.

Port A (Pins 6/7, 12/13): Connect to Processor USB 3.x SS_TX_P/N and SS_RX_P/N. Place 100 nF AC-coupling capacitors on the TX differential lines.

Port B (Pins 17/18, 19/20): Connect to USB-C receptacle TX1_P/N and RX1_P/N.

Port C (Pins 2/3, 21/22): Connect to USB-C receptacle TX2_P/N and RX2_P/N.

Verification: Check for 100 nF DC-blocking capacitors on SuperSpeed TX lines before routing to the host processor.

4. TS5USBA224 (USB 2.0 / Negative-Rail Analog Audio Switch)

VCC (Pin 10): Connect to 3.3V V_{SYS} . Decouple with 0.1 μ F and 1 μ F ceramic capacitors in parallel near the pin.

GND (Pin 5): Connect to Ground plane.

COM1 / COM2 (Pins 1 & 3): Connect to USB-C receptacle D+ and D-.

NC1 / NC2 (Pins 2 & 4 - Data Path): Connect to Processor USB 2.0 PHY (USB_DP and USB_DM).

NO1 / NO2 (Pins 8 & 9 - Audio Path): Connect to Audio DAC / CODEC analog output (AUDIO_LEFT and AUDIO_RIGHT).

SEL / ASEL (Pin 6): Driven by Processor GPIO or TUSB320.OUT1 (0 = \text{USB Data}, 1 = \text{Audio Out}).

Verification: Probe COM1 signal on an oscilloscope when switching SEL; verify negative-going audio sine waves pass without signal clipping down to -2.7V.

5. MAX77751 (3.15A Buck Charger with Integrated Power Path)

CHGIN: Connect directly to USB-C VBUS. Place a 10 μ F (25V rated X7R) ceramic cap + 0.1 μ F cap close to input.

SYS: Main System Supply output rail (powers downstream LDOs, Audio CODEC, and logic ICs).

Place a 22 μ F ceramic cap.

LX (Switching Node): Connect a 1.0 μ H power inductor (rated for $\geq 4\text{A}$ saturation current) between LX and SYS.

BST (Boot Pin): Connect a 0.1 μ F ceramic capacitor directly between BST and LX.

BATT: Connect to the Li-Ion Battery positive terminal (+). Decouple with a 10 μ F capacitor.

CC1 / CC2: Connect in parallel to USB-C CC1 and CC2 (used for charger type detection).

THM: Connect to a 10 k Ω NTC thermistor attached to the battery pack to GND for JEITA temperature monitoring.

EP (Exposed Pad): Connect directly to system Ground using at least 4 to 6 thermal vias.

Verification: Plug in a 5V USB power source and verify SYS output regulates cleanly to 3.8V–4.2V before enabling battery charging.

TUSB320

This is not using instead tusb322 it support usb type c 3.1 right?

Yes, TUSB322 fully supports USB Type-C and USB 3.1 / 3.2 SuperSpeed specifications.

Because CC controllers do not pass high-speed data lines directly, the TUSB322 manages cable orientation, role negotiation (DFP/UFP/DRP), and power detection, while its dedicated DIR output pin drives the HD3SS3212 switch to physically steer the 10 Gbps USB 3.1 SuperSpeed differential data pairs. TUSB322 is actually an upgrade over TUSB320 for audio accessories because it includes built-in VCONN power switching for active audio adapters.

Key TUSB322 Pin Connections for Your Schematic

1

Connect CC Lines

CC1 & CC2 Pins

Connect CC1 and CC2 directly to USB Type-C connector pins A5 and B5. Place small ESD/bypass capacitors (e.g., 220 pF) near the receptacle pins.

2

Connect Cable Orientation Control

DIR Pin

Connect the DIR output pin directly to HD3SS3212.SEL. TUSB322 automatically sets DIR LOW or HIGH depending on cable insertion orientation, controlling the HD3SS3212 to align the USB 3.1 SuperSpeed TX/RX pairs.

3

Wire VBUS Detection

VBUS_DET Pin

Connect VBUS_DET to V_{BUS} through a 900\text{ k}\Omega resistor. TUSB322 uses this input to sense host attachment and initiate role handshake.

4

Supply VCONN Power

VCONN Pin

Connect VCONN to a 5V supply rail. TUSB322 will route this power line to the unassigned CC pin when an active cable or powered Type-C audio accessory is attached.

5

Set Controller Enable & Audio Trigger

EN_N & OUT1/OUT2 Pins

Tie EN_N to GND to keep the IC enabled. Use the output status pins (OUT1/OUT2 or INT_N) to drive TS5USBA224.SEL so the D+/D- lines automatically switch to analog audio when an Audio Adapter Accessory is detected.

Verification: Plug in a Type-C cable, measure the voltage on TUSB322.DIR with a multimeter, flip the cable 180 degrees, and verify the DIR pin toggles state (0V to 5V).

How do I configure TUSB322 in GPIO mode to automatically switch TS5USBA224 to audio mode without an external MCU?

To configure the TUSB322 in hardware GPIO mode for automatic analog audio switching without a microcontroller (MCU), set the TUSB322 logic mode, map its Audio Accessory output pin, and invert the control signal for the TS5USBA224.

Logic Pin Configuration & Hardware Setup

Set TUSB322 to GPIO Mode

ADDR Pin (Pin 11): Leave floating / Not Connected (NC). TUSB322 samples ADDR at power-up; leaving it floating forces the IC into GPIO mode instead of I2C mode.

EN_N Pin: Tie directly to GND to keep the controller permanently enabled.

PORT Pin: Connect to VDD via a 10 kΩ pull-up resistor for Dual Role Port (DRP), or ground it if operating strictly as an Upstream Facing Port (UFP / Device).

Audio Accessory Signal Line (OUT3)

In GPIO mode, the multi-function INT_N / OUT3 pin acts as OUT3 (Audio Accessory Detect).

No Audio Accessory Attached: OUT3 remains HIGH (3.3V / 5V pulled up).

Audio Accessory Attached: OUT3 pulls LOW (0V) when an audio adapter dongle (R_a / R_a or R_a / R_d termination) is detected on the CC lines.

Signal Inversion for TS5USBA224 (ASEL)

The TS5USBA224 audio switch requires ASEL = HIGH to route the analog audio path (NO1/NO2) to D+/D-, and ASEL = LOW to select USB 2.0 data (NC1/NC2).

Because TUSB322.OUT3 outputs LOW when audio is detected, insert a single-gate logic inverter (e.g., SN74LVC1G04 or SN74AUP1G04) between TUSB322.OUT3 and TS5USBA224.ASEL.

1

Configure TUSB322 Control Pins

Hardware Setup

Leave ADDR pin floating (NC) to lock TUSB322 in GPIO mode. Connect EN_N to GND and pull PORT to VDD via 10 kΩ resistor.

2

Place Logic Inverter Gate

Signal Polarity Alignment

Connect TUSB322 pin 11 (OUT3) to the input (A) of an SN74LVC1G04 single-gate inverter.

Connect a 10 kΩ pull-up resistor from OUT3 to V_{DDIO} to ensure a stable HIGH logic level when floating.

3

Route Inverter Output to Switch

TS5USBA224 Interconnect

Connect the inverter output (Y) directly to TS5USBA224.ASEL (Pin 6). When an audio accessory is plugged in, OUT3 drops LOW, turning the inverter output HIGH, which flips TS5USBA224 to Audio Mode automatically.

Verification: Plug a USB-C to 3.5mm headphone adapter into the receptacle and measure the voltage on TS5USBA224.ASEL with a multimeter; verify it jumps from 0V up to logic HIGH (≥ 1.8V) automatically upon insertion.

Phone's single usb type c port is doing 3 things

That single USB Type-C port acts as a multi-purpose interface by dynamically re-routing its physical pins into three distinct operating hardware paths based on what cable or accessory is plugged in.

ModeActive USB-C PinsRoute DestinationPrimary Controller / Switch IC

1. ChargingVBUS, GND, CC1, CC2Power Path & ChargerMAX77751 (manages power path & charges battery up to 3.15A)

2. Data TransferD+/D- (USB 2.0)

TX1/2, RX1/2 (USB 3.x)Host Processor PHYTS5USBA224 (routes D+/D-)

HD3SS3212 (routes SuperSpeed TX/RX)

3. Analog AudioD+ (Right Ch), D- (Left Ch)

SBU1/SBU2 (Mic/GND)Audio DAC / Headphone Amp

How Isolation Is Maintained

Detection Phase: When nothing is plugged in, TUSB322 monitors the CC pins (CC1/CC2).

Charger / Data Mode: When a normal USB host or charger is plugged in, TUSB322 leaves TS5USBA224 in Data Mode (connecting D+/D- to the processor) and sets HD3SS3212 orientation. MAX77751 draws power from VBUS.

Audio Mode: When an analog audio adapter is plugged in, TUSB322 detects R_a / R_d termination resistance on the CC lines, pulls OUT3 LOW, and flips TS5USBA224 to route D+/D- directly to the DAC's analog audio lines while disconnecting the processor's USB 2.0 PHY to prevent short circuits.